

SMART WATER BOTLLE

Introduction

Hydration is the process of providing the body with an adequate amount of water to maintain normal bodily functions. Proper hydration is important for maintaining body temperature, supporting digestion, transporting nutrients, and ensuring that the body functions effectively. However, many people do not drink enough water throughout the day, often because they are busy or simply forget to drink water regularly.

The **Smart Water Bottle** is an embedded system designed to help users monitor and improve their daily water consumption. The system will measure the amount of water in the bottle using a **load cell** and process the sensor data using a **microcontroller**. By monitoring changes in the weight of the bottle, the system can estimate the amount of water consumed by the user.

When the system detects that the user has not consumed enough water within a specified period, it will provide audible and visual reminders to encourage regular hydration. The project combines sensors, microcontroller programming, data processing, and output devices to create a practical embedded system that promotes healthy hydration habits.

2. Functional Requirements

The system shall:

FR1 — Monitor Water Quantity

Measure the weight of the water bottle using a **load cell**.

Continuously or periodically detect changes in weight.

Calculate the amount of water consumed based on the change in measured weight.

FR2 — Track Water Consumption

Record the initial amount of water in the bottle.

Detect when the bottle's weight decreases.

Calculate the amount of water consumed.

Example:

Initial water weight: 1000 g

Current water weight: 750 g

Water consumed: 250 mL

Since approximately **1 gram of water = 1 milliliter**, the system can use weight change to estimate water consumption.

FR3 — Detect Drinking Activity

Detect a meaningful reduction in bottle weight.

Ignore very small changes caused by sensor noise or movement.

Update the total amount of water consumed when a valid drinking event is detected.

FR4 — Hydration Reminder

Remind the user to drink water if no drinking activity is detected for a predefined period.

For example:

No water consumed for 60 minutes → Reminder activated.

FR5 — LED Notification

Activate an LED when a hydration reminder is triggered.

Turn the LED off automatically after **3 seconds** to avoid continuously shining into the user's eyes.

FR6 — Audible Notification

Activate a buzzer together with, or as an alternative to, the LED reminder.

Automatically stop the buzzer after a predefined duration.

FR7 — Reset Drinking Timer

When the system detects that the user has consumed water, reset the inactivity timer.

Start monitoring again for the next hydration interval.

FR8 — Sensor Data Processing

The microcontroller shall:

Receive weight measurements from the load cell.

Use the **HX711 amplifier/ADC** to convert the load cell's small analog signal into a digital signal.

Filter or average readings to reduce noise.

Determine whether a weight change represents actual water consumption.

3. Hardware Requirements

The initial prototype will include:

Component	Purpose
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Arduino	Main controller that processes sensor data and controls outputs
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Load Cell	Measures the weight of the bottle and water
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HX711	Amplifies and digitizes the load cell signal for the Arduino
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LED	Provides a visual hydration reminder
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Buzzer	Provides an audible hydration reminder
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Breadboard	Used for prototype circuit connections
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Jumper Wires	Connect components
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Power Source	Supplies power to the system
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4. Software Requirements

The firmware shall:

Read data from the HX711.

Convert sensor readings into weight measurements.

Calibrate the load cell.

Filter noisy sensor readings.

Calculate water consumption.

Track the time since the last drinking event.

Trigger LED and/or buzzer reminders.

Automatically deactivate the LED after 3 seconds.

Reset the reminder timer when drinking is detected.

5. Non-Functional Requirements

NFR1 — Accuracy

The system should reasonably detect changes in water quantity. Exact medical-grade accuracy is not required for the prototype.

NFR2 — Reliability

The system should avoid false drinking detections caused by small sensor fluctuations.

NFR3 — Safety

The electronic components should remain separate from the water storage area and be protected from water exposure.

NFR4 — Power Efficiency

The system should avoid unnecessarily keeping LEDs or buzzers active for long periods.

NFR5 — Modularity

The hardware and software should be designed so future features can easily be added, such as:

Bluetooth connectivity

Mobile application integration

Daily hydration goals

LCD/OLED display

Battery-powered operation

Consumption history tracking

6. Minimum Viable Product (MVP)

For your **first version**, I recommend we keep it simple:

The system measures the bottle's weight using a load cell, detects when water has been consumed, tracks the amount consumed, and activates an LED reminder for 3 seconds when no drinking activity has been detected for a specified period.